EXP12 (a)	End - End Communication at Transport Layer Implement echo client server using TCP/UDP

Aim:

To simulate end - end communication at Transport layer to implement echo client server using TCP/UDP.

TCP:

```
import socket
import threading
def start - server():
    server = socket.socket(socket.AF_INET)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000
    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")

    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower()
        print("[SERVER] Connection closed")
        break
    print(f"[SERVER] Received : {data}")
    conn.send(f"Server received : {data}".
                encode())
```

```
conn.close()
server.close()
```

Client :

```
def start_client():
    client = socket.socket(socket.AF_INET,
                           socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")

    while True:
        msg = input("Enter message (or 'quit' to
                    exit): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")

    client.close()
```

Main :

```
if __name__ == "__main__":
    server_thread = threading.Thread(target=
                    start_server, True)
    server_thread.start()
    start_client()
```

Output:-

[SERVER] Listening on 127.0.0.1 :5000 [CLIENT]
connected to server.

[SERVER] Connected by ('127.0.0.1:5000 [CLIENT]
Enter message '(or 'quit' to exit):

hello

[SERVER] Received: hello

[SERVER RESPONSE] : Server received : hello
Enter message [or 'quit' to exit):

[SERVER] Listening on 127.0.0.1.5000 [CCLIENT]
Connected to server

[SERVER] Connected by ('127.0.0.1, 60328)
Enter message (or 'quit' to exit):

Hi

[SERVER] Received : Hi

[SERVER] Connection closed by client

Result:- Thus implementation of echo client server using TCP/UDP done successfully.